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Statistics & Econometrics

Exam: Statistics & Econometrics / 2025-Q2 Exam and Retake
Examiner: Prof. Dr. Emanuel Mönch
 Prof. Dr. Thilo Kind

Date: Thursday 18th December, 2025
Time: 10:00 – 11:20

Working instructions

- This exam consists of **20 pages** with a total of **11 questions**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable points in this exam is 70 credits.
- **Detaching pages from the exam is prohibited.**
- Allowed resources:
 - any **paper-based materials** (slides, textbook, formula sheet)
 - one **FS approved non-programmable calculator**
- **Answers to quantitative questions are only accepted if the solution approach is documented.**
Give a reason for each quantitative question in the respective solution box below.

Left room from _____ to _____ / Early submission at _____



☐ Exam empty





Dear MoF Econometrics students,

This is the final exam for our class. The exam will last 80 minutes (which includes reading time) and will be worth 70 performance points, out of the total 120 for the course.

This is a pen & paper exam. You can use a formula sheet, text books or other printed-out materials. There is no restriction on the number of pages of the supporting materials. Only non-programmable calculators from the list of FS permissible calculators are allowed.

You will not be allowed to communicate with each other during the assignment. Your mobile phones and other electronic devices will have to be stowed away.

If you struggle with one exercise, move on to the next ones in order to not lose time.

There are solution boxes after some questions where you shall enter your reasoning for the corresponding question. **Please ensure to strike out invlaid solutions.**

Please note that the language for the exam is English. Therefore, a "." is a decimal and a "," is a thousand separator. For example, $4/5 = 0.8$ and NOT 0,8. As another example, $100 \cdot 100 = 10,000$ and NOT 10.000.

Mark correct answers with a cross



To undo a cross, completely fill out the answer option



To re-mark an option, use a human-readable marking



Good luck!





Problem 1 Test for Serial Correlation (2 credits)

One of the following statements is correct. Mark the CORRECT statement.

- ☐ The Goldfeld-Quandt test can be used to detect serial correlation in the residuals of a linear regression model.
- ☐ The Dickey-Fuller test can be used to detect serial correlation in the residuals of a linear regression model.
- ☐ The Durbin-Watson test can be used to detect serial correlation in the residuals of a linear regression model.
- ☐ The Newey-West test can be used to detect serial correlation in the residuals of a linear regression model.

Problem 2 Intercept in a Linear Regression Model (2 credits)

One of the following statements is correct. Mark the CORRECT statement.

- ☐ In a linear regression model with an intercept the sum of squared residuals is guaranteed to be zero on average.
- ☐ In a linear regression model without an intercept the residuals will be serially correlated.
- ☐ In a linear regression model without an intercept the residuals are not guaranteed to be zero on average.
- ☐ In a linear regression model with an intercept the R-squared can be negative.

Problem 3 Biases (2 credits)

One of the following statements is correct. Mark the CORRECT statement.

- ☐ Measurement error in the dependent variable mitigates the estimated relationship between the dependent and the independent variable.
- ☐ The slope coefficient in a linear regression without an intercept will be unbiased.
- ☐ Omitted variable bias can mitigate the estimated relationship between the dependent and the independent variable.
- ☐ Measurement error in the independent variable leads to an overestimation of the slope coefficient.





Problem 4 Recession Prediction (8 credits)

Stock prices tend to fall sharply ahead of recessions. That said, not all bear markets are followed by a recession. Nobel laureate Paul Samuelson once famously quipped: "The stock market has predicted nine of the last five recessions!"

Suppose you are an analyst aiming to assess the probability of an impending recession using the stock market as a leading indicator. You observe a recession indicator variable REC_t which assumes the value of 1 if period t falls in a recession and 0 otherwise. Moreover, you download the annual log return on the stock market index, $RetMkt_t = \ln Mkt_t - \ln Mkt_{t-12}$. Given these two inputs, you would like to test whether a decline in the annual market return signals a recession starting in three months time.

a) Which regression model do you need to estimate to analyze this relationship? Mark the CORRECT answer. (2 credits)

- ☐ A panel model relating REC_{t+3} to $RetMkt_t$.
- ☐ A Logit model or a Probit model relating REC_{t+3} to $RetMkt_t$, both would give similar results.
- ☐ It would be logical to estimate a Logit model because probably the Probit model is not the right approach.
- ☐ A Probit model relating REC_{t+3} to $RetMkt_t$.
- ☐ A linear regression model regressing $RetMkt_{t+3}$ on REC_t .

Suppose you end up estimating the following probit model

$$\begin{aligned} P(REC_{t+3} = 1 | RetMkt_t) &= \Phi(\hat{\alpha} + \hat{\beta} RetMkt_t) \\ &= \Phi(-0.1 - 0.3 \cdot RetMkt_t) \end{aligned}$$

with both estimates being highly statistically significant.

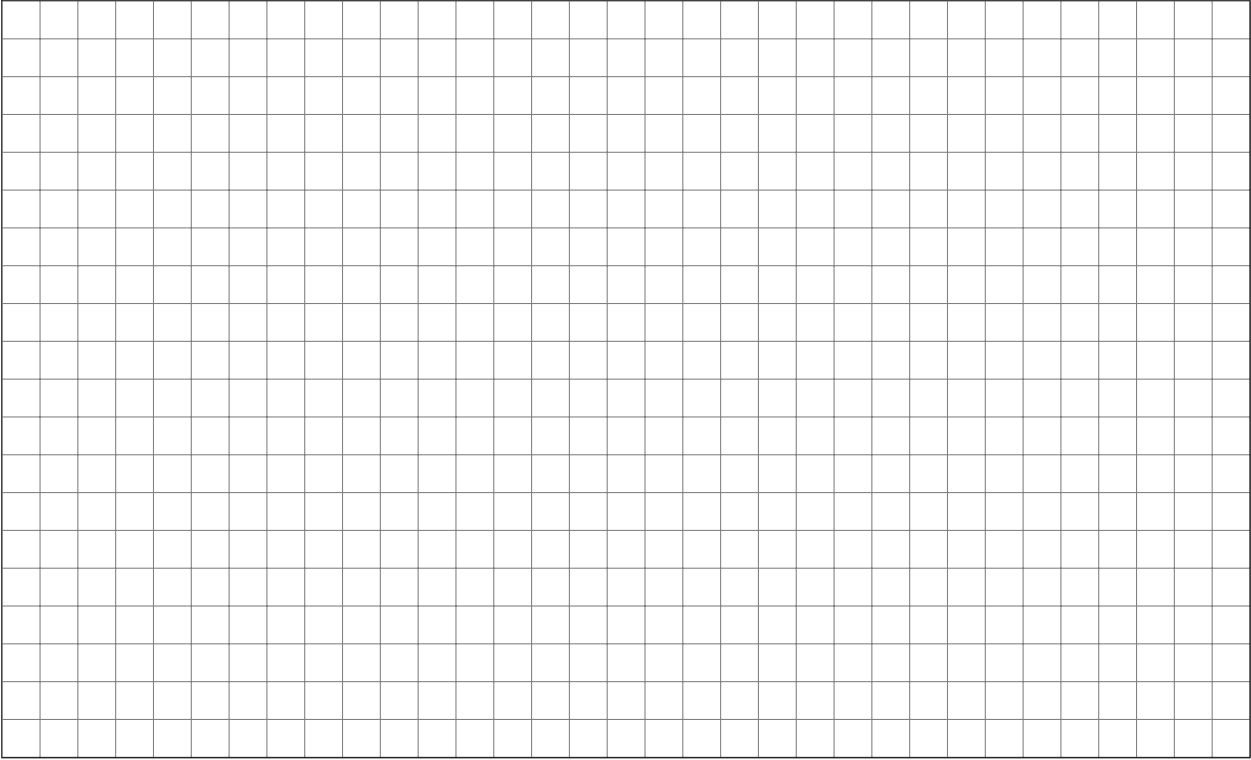
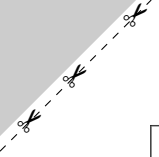
b) Mark the CORRECT statement regarding the coefficients $\hat{\alpha}$ or $\hat{\beta}$. (3 credits)

- ☐ The coefficient $\hat{\alpha}$ measures the average probability of recessions irrespective of the market return. A negative $\hat{\alpha}$ implies that the average probability of a recession is below 50 percent.
- ☐ The coefficient $\hat{\beta}$ measures the sensitivity of the recession probability to movements in the market return. The estimated coefficient suggests that the probability of recession decreases by 3 percent if the return on the market increases by 10 percent.
- ☐ The negative estimate $\hat{\beta}$ suggests that the probability of a recession increases when the return on the market declines.
- ☐ The coefficient $\hat{\alpha}$ measures the sensitivity of the recession probability to movements in the market return. The estimated coefficient suggests that the probability of recession decreases by 1 percent if the return on the market increases by 10 percent.
- ☐ The coefficient $\hat{\alpha}$ measures the sensitivity of the recession probability to movements in the market return. The estimated coefficient suggests that the probability of recession increases by 1 percent if the return on the market increases by 10 percent.

c) Suppose you would call a recession when the model-implied probability of a recession rises above 50%. Mark the CORRECT statement about when you would call a recession given the estimates $\hat{\alpha} = -0.1$ and $\hat{\beta} = -0.3$. (3 credits)

- ☐ The annual return on the market $RetMkt$ would have to drop below 20 percentage points.
- ☐ The annual return on the market $RetMkt$ would have to drop below negative 45 percentage points.
- ☐ The annual return on the market $RetMkt$ would have to drop below negative 33 percentage points.
- ☐ The annual return on the market $RetMkt$ would have to drop below negative 20 percentage points.
- ☐ The annual return on the market $RetMkt$ would have to drop below 33 percentage points.
- ☐ The annual return on the market $RetMkt$ would have to drop below 45 percentage points.







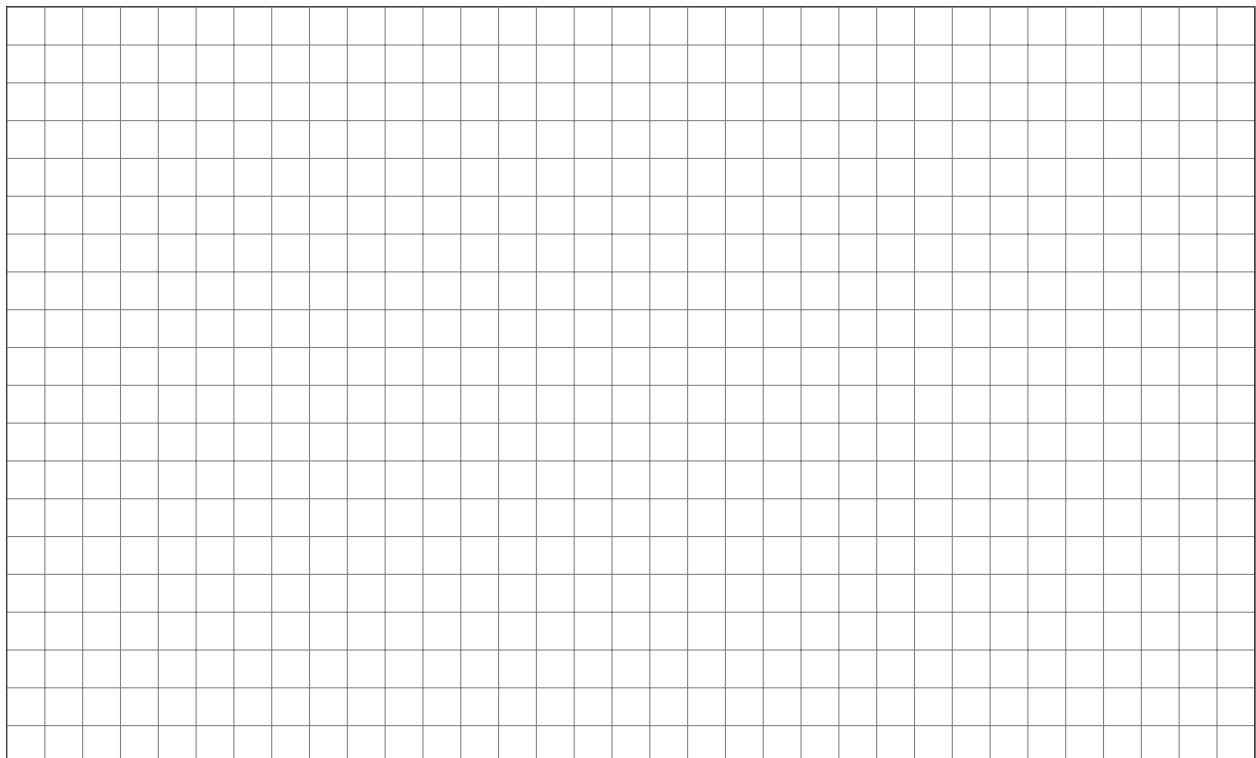
Problem 5 Confidence Interval (4 credits)

Assume you are given the following results for a univariate regression of an excess stock return on the excess market return with $T = 2000$ observations:

$$\begin{aligned} r_{it} &= \hat{\alpha} + \hat{\beta} r_{mt} + \hat{u}_t \\ &= 5.0 + 2.0 r_{mt} + \hat{u}_t \\ &\quad (4.0)(0.8) \end{aligned}$$

The standard errors are given in parentheses below the coefficient estimates. Based on these regression results, what is the lower bound of the two-sided 90% confidence interval for β ? The standard normal critical values for two-tailed tests with 99%, 95%, and 90% significance level equal 2.326, 1.960 and 1.645, respectively.

- ☐ -3.85
- ☐ 0.68
- ☐ -3.12
- ☐ 4.09
- ☐ 0.14
- ☐ 0.43
- ☐ 3.32
- ☐ -2.49





Problem 7 Hedge Portfolio (5 credits)

Assume that you are a portfolio manager in a quantitative asset management company. A client asks you to build a portfolio of tech firms that is not exposed to overall market risk. Your team has put together such a portfolio (called TMM, “tech-minus-market”) that eliminates market risk. They document this by showing you a slide deck that includes the results of a regression of the TMM portfolio’s monthly return (in excess of the risk-free rate) on the excess market return over the last ten years . These are the results:

$$\begin{aligned} r_{TMM,t} &= 0.9 + 0 \cdot r_{MKT,t} + \hat{u}_t \\ &= (0.2) \quad (0.8) \end{aligned}$$

The slide deck also provides the following pieces of information. Over the last ten years:

- i) The correlation of the excess return on TMM and the NASDAQ, a common index for the tech industry, was 80 percent or 0.8.
- ii) The monthly excess return of the NASDAQ over the risk-free rate was 0.5 percent.
- iii) The monthly sample standard deviation of TMM’s excess return was 12 percent.
- iv) The monthly sample standard deviation of the NASDAQ’s excess return was 8 percent.

a) What is the average monthly TMM portfolio return in excess of the risk-free rate? (1 credit)

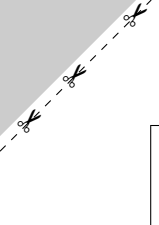
- ☐ $\hat{\alpha} = 0.1\%$
- ☐ $\hat{\alpha} = 0.8\%$
- ☐ $\hat{\alpha} = 0.6\%$
- ☐ $\hat{\alpha} = 0.3\%$
- ☐ $\hat{\alpha} = 0.9\%$
- ☐ $\hat{\alpha} = 0\%$

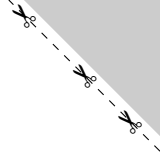
On a Friday afternoon your client calls and wants to know where you stand with the portfolio. You tell her that you essentially have it all ready but still need to check a few things. The client is impatient and wants to know what the “alpha” of the portfolio is relative to the NASDAQ. You tell the client that you need a few minutes to answer this question.

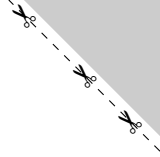
b) Unfortunately, you realize that your entire team has already left for the weekend. What’s worse, you do not have the portfolio return data to check this yourself. Luckily, you learned a few things in your econometrics class at Frankfurt School. After a few minutes you call back the client and tell her that the portfolio’s monthly alpha with respect to the NASDAQ over the last ten years was equal to (4 credits)

- ☐ $\hat{\alpha} = 0.5\%$
- ☐ $\hat{\alpha} = 0.4\%$
- ☐ $\hat{\alpha} = 0.3\%$
- ☐ $\hat{\alpha} = 0.6\%$
- ☐ $\hat{\alpha} = 0.8\%$
- ☐ $\hat{\alpha} = 0.9\%$
- ☐ $\hat{\alpha} = 0.7\%$
- ☐ $\hat{\alpha} = 0.2\%$








$$P(y_i = 1|x_i) = L(\hat{\alpha} + \hat{\beta}x_i)$$

□ $P(y_i = 1) = 99.9\%$

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□ $ME = 0.6\%$

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Problem 11 Time Series (22 credits)

Remark: Each sub-question is independent and can be answered separately.

You advise an investment committee on the monthly EURUSD log-return Y_t . First, consider an MA(1) process following

$$Y_t = \mu + u_t + \theta u_{t-1}, \quad \mu = 1.0, \theta = 0.2, u_t \sim \mathcal{N}(0, \sigma_u^2), \sigma_u^2 = 0.1$$

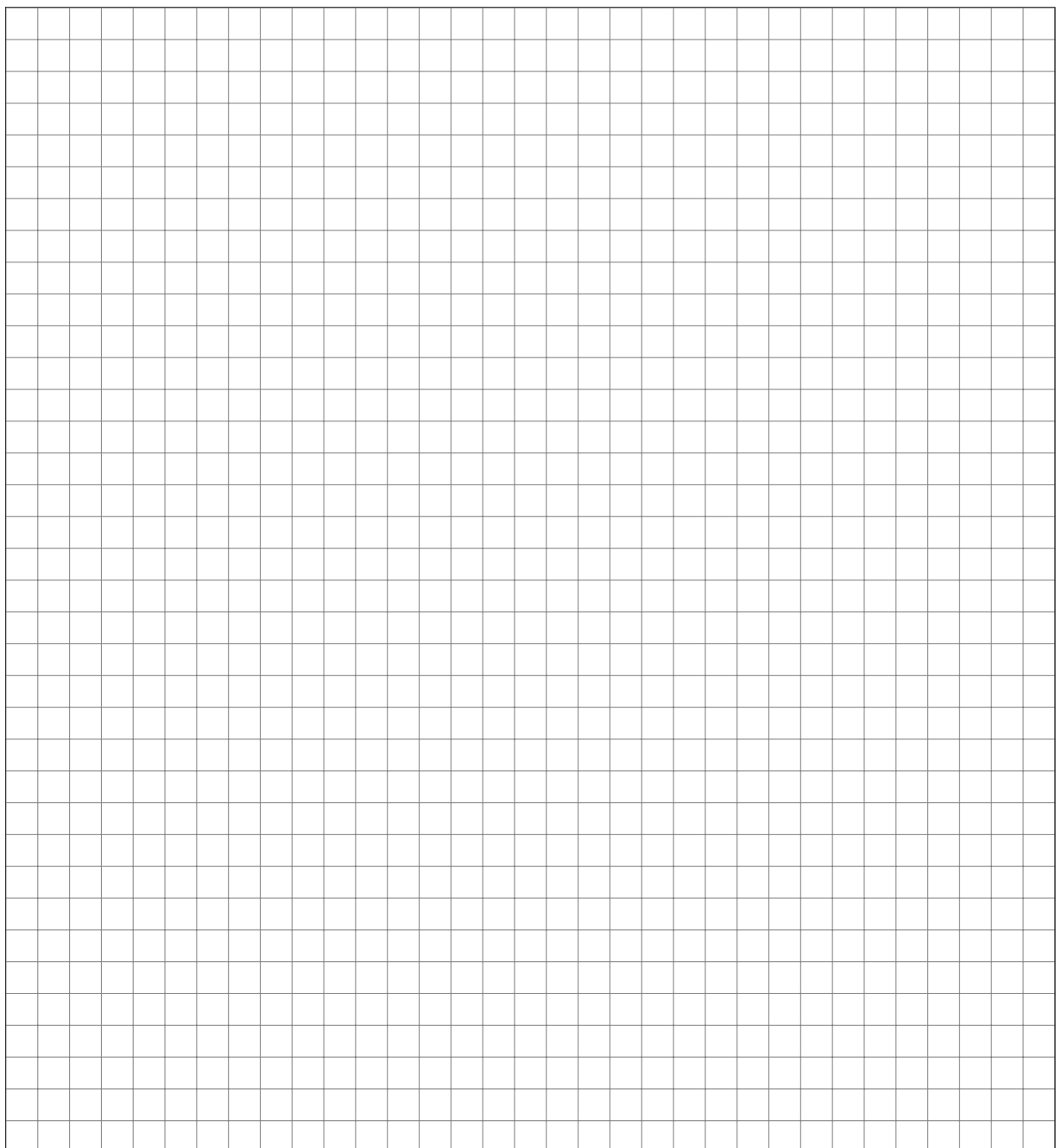
a) Compute the autocovariance $\gamma_1 = \text{Cov}(Y_t, Y_{t-1})$ and the autocorrelation $\rho_1 = \text{Cov}(Y_t, Y_{t-1}) / \text{Var}(Y_t)$ (4 credits)

☐ $\gamma_1 = 0.02, \rho_1 \approx 0.020$

☐ $\gamma_1 = 0.01, \rho_1 \approx 0.095$

☐ $\gamma_1 = 0, \rho_1 = 0$

☐ $\gamma_1 = 0.02, \rho_1 \approx 0.192$





b) You then turn to the data and inspect the autocorrelation function plot (ACF) and the partial autocorrelation function plot (PACF). The ACF declines geometrically, the PACF cuts off to zero after lag 1. Which specification is most consistent with that evidence? (2 credits)

- ☐ *ARMA*(1, 1)
- ☐ *AR*(1)
- ☐ Random walk
- ☐ *GARCH*(1, 1)
- ☐ *MA*(1)

c) You want to evaluate different Autoregressive models $AR(p)$ for orders $p = 0, \dots, 9$ for the 1-D data stored in `df`. Fill in the blank: (2 credits)

```
from statsmodels.tsa.arima.model import ARIMA
```

```
res = {}  
for p in range(10):  
    mod = _____  
    res[p] = mod.fit()
```

- ☐ `ARIMA(df, order=(p,0,0)).fit()`
- ☐ `ARIMA(df, order=(p,0,0))`
- ☐ `ARMA(df, order=(p,0,0))`
- ☐ `ARIMA(df, order=(0,0,p))`
- ☐ `ARIMA(res, order=(p,0,0))`
- ☐ `model.AR(df, order=p)`



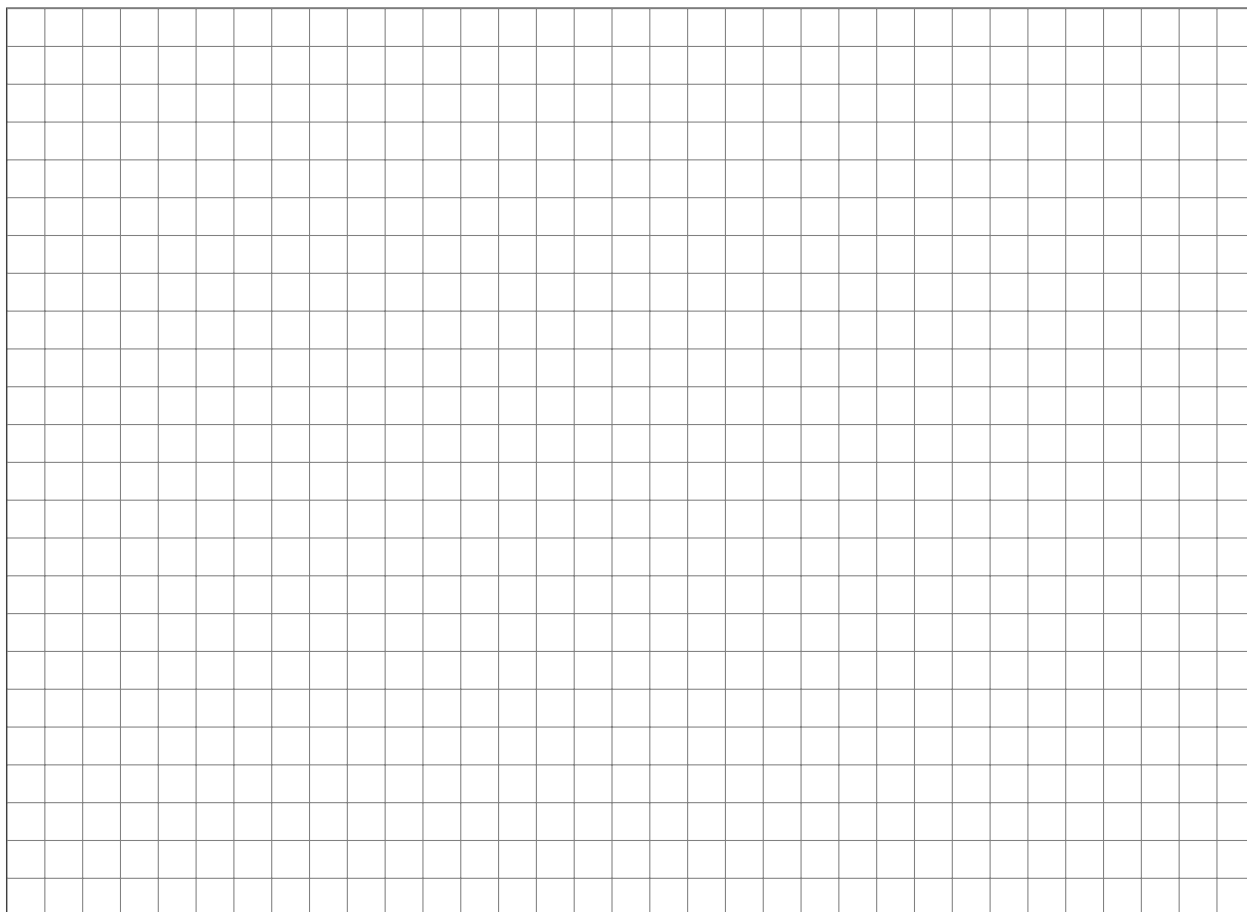


d) You decide to continue with an $AR(1)$ and obtain the fit

$$Y_t = 1.0 + 0.95Y_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_u^2), \quad \sigma_u^2 = 0.2$$

Compute the unconditional volatility $Vol(Y_t) = \sqrt{Var(Y_t)}$ (4 credits)

- ☐ $Vol(Y_t) = 20.0$
- ☐ $Vol(Y_t) \approx 2.051$
- ☐ $Vol(Y_t) \approx 1.432$
- ☐ $Vol(Y_t) = 0.2$



e) Concerned about risk, you test for conditional heteroskedasticity. The auxiliary regression on monthly residual squares with 5 lags yields an $R^2 = 6\%$ with $T = 240$ periods. What is the LM statistic and the conclusion at 5% significance, given that $\chi_{0.95}^2 = 11.07$? (3 credits)

- ☐ 7.2, do not reject
- ☐ 14.4, do reject
- ☐ 14.4, do not reject
- ☐ 1440, do reject
- ☐ 7.2, do reject
- ☐ 1440, do not reject





f) Given the evidence, you therefore estimate an AR(1)-ARCH(1) model:

$$Y_t = 1.0 + 0.95Y_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_t^2)$$

$$\sigma_t^2 = 0.2 + 0.9u_{t-1}^2$$

Compared to the constant-variance case $\sigma_u^2 = 0.2$ above, does the unconditional variance of the AR(1)-ARCH(1) model ($E[\sigma_t^2]$) ... (3 credits)

- ☐ Increase
- ☐ Decrease
- ☐ Stay the same
- ☐ Become non-stationary

g) Complete the code below for a monthly rolling window of length 60 by filling in the blank (2 credits)

```
from sklearn.linear_model import LinearRegression

train_window_months = 60

y_all = df['Return_1(+1M)']
X_all = df[['DivYld']]

res = {}

for t in range(_____, len(df)-train_window_months):
    X_train = X_all[t : t + train_window_months]
    y_train = y_all[t : t + train_window_months]

    model = LinearRegression()
    res[t] = model.fit(X_train, y_train)
```

- ☐ X_all
- ☐ 1
- ☐ y_all
- ☐ 0
- ☐ t
- ☐ train_window_months

h) What concern does a pseudo-out-of-sample exercise most likely address? (2 credits)

- ☐ Serial correlation
- ☐ Overfitting and look-ahead bias
- ☐ Measurement error
- ☐ Heteroskedasticity





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